

Exploratory Test Report

dataghurhi-auth

Report generated	2026-09-11 19:06 UTC
Run window	2026-09-11T18:45:55.393029+00:00 to 2026-09-11T19:06:17.353347+00:00
Test cases executed	5
Defects found	1
Prepared by	Exploratory Testing Planner Agent (automated)

Every outcome in this report is attributed by the test system itself. A failed run is counted as a defect only when the application was at fault; runs lost to the limits of the automated tester or to the test environment are reported separately and are not defects.

Executive Summary

A total of five test runs were executed against the dataghurhi-auth application, comprising 169 steps over approximately 889 seconds. One run was informative and produced a defect finding, while three runs were lost to automated tester limitations and one was blocked by an environmental guardrail. The single application defect identified was an assertion failure in the survey creation validation flow, where a whitespace-only title was expected to be rejected. The most important takeaway is that while one validation defect was confirmed, the majority of planned coverage could not be completed due to tool navigation failures and a captcha block.

Scope and Method

The tests set out to verify validation behavior for survey creation, analysis type handling, statistical procedure type mismatches, expired survey response handling, and preprocessing cell edits. Each test was executed as an automated browser navigation sequence against the live application at `dataghurhi.cse.buet.ac.bd`, with the system recording step counts, navigation paths, and outcome attributions.

Why These Tests Were Necessary

The test set targeted input validation and data type enforcement, which are critical to the reliability of a data-oriented survey and analysis platform. The tests covered both creation-time and runtime validation, including whitespace handling, numerical column requirements, unsupported data type rejection, expired survey behavior, and cell edit type checks. These areas are likely to reveal user-facing errors that would undermine data integrity or cause silent failures. The inclusion of a captcha-blocked run also confirmed that the application enforces external security controls during survey submission.

What the Tests Examined

The planned tests checked that a survey with a whitespace-only title is rejected, that an analysis requiring a numerical column refuses to run when only categorical columns are selected with a message naming the missing column, that the analysis module refuses to run a statistical procedure on a column of an unsupported data type without coercion or value dropping, that a survey with an expired ending date refuses new responses with an explicit reason, and that the preprocessing module rejects a cell edit whose value does not match the column's data type.

Discussion of Findings

One defect was found in the application. The test titled "Verify survey creation with whitespace-only title is rejected" resulted in an assertion failure, indicating that the application did not behave as expected when a whitespace-only title was submitted. The failure occurred at the application's base URL. No other defects were identified in the completed runs.

Limitations of This Run

Three runs were lost to navigation livelocks, a limitation of the automated tester where the tool repeatedly navigated between pages without reaching the target state; these were not defects in the application. One run was blocked by a captcha guardrail, which prevented the tester from proceeding past a survey submission step. These four runs did not produce application findings and should not be interpreted as evidence of application behavior.

Recommendation

The application defect concerning whitespace-only survey titles should be reproduced manually and corrected so that such input is rejected with an appropriate validation message. The automated tester should be reconfigured to avoid navigation livelocks, for example by adding explicit waits or more direct routing to target pages. The captcha-blocked test should be rerun using a method that satisfies the guardrail, such as providing valid credentials or a test bypass. After these steps, the four incomplete tests should be re-executed to obtain full coverage of the planned validation scenarios.

Results by Attribution

OUTCOME	N	SHARE	MEANING
Passed	0	0%	the application behaved as expected
Defect	1	20%	a fault in the application under test
Tool limit	3	60%	our automated tester could not complete the run
Blocked	1	20%	test data, permissions, guardrail or step budget

Total execution effort: 169 steps across 889 seconds. Of 5 runs, 1 produced evidence about the application; the remainder were lost to the tester or the environment and say nothing about its quality.

Knowledge coverage: 5 test cases recorded, 5 of 71 requirements covered, 0 user-interface states mapped.

Tests Executed

1. Verify survey creation with whitespace-only title is rejected

Defect 8 steps, 32s, recorded as ASSERTION_FAILURE

Path: <https://dataghurhi.cse.buet.ac.bd/>

2. Verify that an analysis requiring a numerical column refuses to run when only categorical columns are selected, with a message naming the missing numerical column

Tool limit 58 steps, 232s, recorded as NAVIGATION_LIVELOCK

Path: <https://dataghurhi.cse.buet.ac.bd/>

<https://dataghurhi.cse.buet.ac.bd/?tab=projectdetails&projectId=268&source=projects>

<https://dataghurhi.cse.buet.ac.bd/?tab=analysis>

<https://dataghurhi.cse.buet.ac.bd/?tab=projectdetails&projectId=268&source=projects>

<https://dataghurhi.cse.buet.ac.bd/?tab=analysis>

<https://dataghurhi.cse.buet.ac.bd/?tab=projectdetails&projectId=268&source=projects>

<https://dataghurhi.cse.buet.ac.bd/view-survey/597> > <https://dataghurhi.cse.buet.ac.bd/>

3. Verify that /analysis refuses to run a statistical procedure on a column of an unsupported data type (e.g., text) with a message naming the mismatch, and does not coerce or drop values

Tool limit 48 steps, 260s, recorded as NAVIGATION_LIVELOCK

Path: <https://dataghurhi.cse.buet.ac.bd/> >
<https://dataghurhi.cse.buet.ac.bd/?tab=projectdetails&projectId=268&source=projects> >
<https://dataghurhi.cse.buet.ac.bd/?tab=analysis> >
<https://dataghurhi.cse.buet.ac.bd/?tab=projectdetails&projectId=262&source=projects> >
<https://dataghurhi.cse.buet.ac.bd/view-survey/594> >
<https://dataghurhi.cse.buet.ac.bd/?tab=projectdetails&projectId=262&source=projects> >
<https://dataghurhi.cse.buet.ac.bd/?tab=analysis> >
<https://dataghurhi.cse.buet.ac.bd/?tab=projectdetails&projectId=262&source=projects>

4. Verify that a survey with an expired ending_date refuses new responses with an explicit reason and does not silently discard the submission

Blocked 21 steps, 232s, recorded as BLOCKED_BY_CAPTCHA

Path: <https://dataghurhi.cse.buet.ac.bd/> >
<https://dataghurhi.cse.buet.ac.bd/?tab=projectdetails&projectId=262&source=projects> >
<https://dataghurhi.cse.buet.ac.bd/?tab=projectdetails&projectId=268&source=projects> >
<https://dataghurhi.cse.buet.ac.bd/view-survey/597> > <https://dataghurhi.cse.buet.ac.bd/v/b39-c02> >
<https://dataghurhi.cse.buet.ac.bd/survey-success> > <https://dataghurhi.cse.buet.ac.bd/> >
<https://dataghurhi.cse.buet.ac.bd/?tab=projects>

5. Verify /preprocess rejects a cell edit whose value does not match the column's data type

Tool limit 34 steps, 133s, recorded as NAVIGATION_LIVELOCK

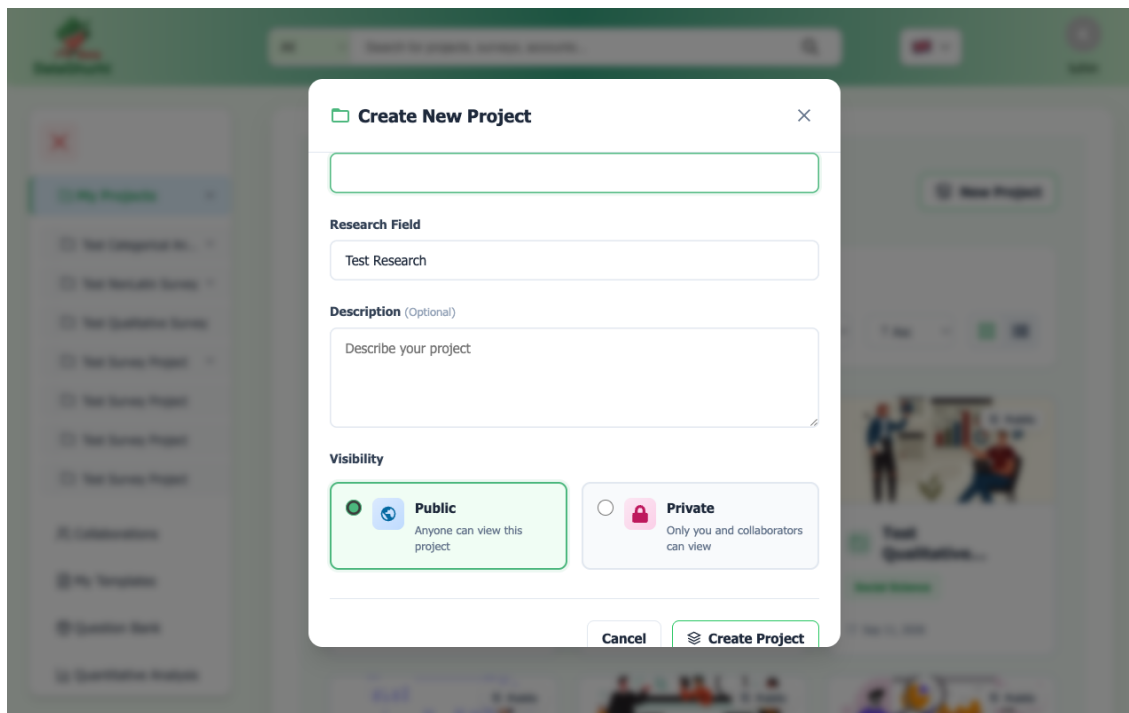
Path: <https://dataghurhi.cse.buet.ac.bd/> >
<https://dataghurhi.cse.buet.ac.bd/?tab=projectdetails&projectId=268&source=projects> >
<https://dataghurhi.cse.buet.ac.bd/?tab=analysis> >
<https://dataghurhi.cse.buet.ac.bd/?tab=projectdetails&projectId=262&source=projects> >
<https://dataghurhi.cse.buet.ac.bd/view-survey/594> > <https://dataghurhi.cse.buet.ac.bd/> >
<https://dataghurhi.cse.buet.ac.bd/?tab=projects> >
<https://dataghurhi.cse.buet.ac.bd/?tab=projectdetails&projectId=278&source=projects>

Defects and Where They Occurred

Verify survey creation with whitespace-only title is rejected

Where: <https://dataghurhi.cse.buet.ac.bd/>

Classified as: ASSERTION_FAILURE (a fault in the application)



Screen captured at the point of failure (TC-001-failed.png).

Runs That Are Not Defects

The runs below did not complete, but the cause lies with the automated tester or the test environment rather than the application. They are excluded from the defect count above and must not be read as quality problems in the software under test.

Verify that an analysis requiring a numerical column refuses to run when only categorical columns are selected, with a message naming the missing numerical column

Tool limit: NAVIGATION_LIVELOCK - our automated tester could not complete the run

Verify that /analysis refuses to run a statistical procedure on a column of an unsupported data type (e.g., text) with a message naming the mismatch, and does not coerce or drop values

Tool limit: NAVIGATION_LIVELOCK - our automated tester could not complete the run

Verify that a survey with an expired ending_date refuses new responses with an explicit reason and does not silently discard the submission

Blocked: BLOCKED_BY_CAPTCHA - test data, permissions, guardrail or step budget

Verify /preprocess rejects a cell edit whose value does not match the column's data type

Tool limit: NAVIGATION_LIVELOCK - our automated tester could not complete the run